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Potential for the Formation of N-Nitrosamines during the Manufacture of Active Pharmaceutical Ingredients: An Assessment of the Risk Posed by Trace Nitrite in Water

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ABSTRACT: Regulatory requests that marketing authorization holders for chemically synthesized active substances risk assess their medicines for the potential presence of N nitrosamines have led to a renewed interest in amine nitrosation. We have used published mechanistic and kinetic studies of amine nitrosation to assess the risk that traces of nitrite in the water used during active pharmaceutical ingredient (API) manufacturing could give rise to significant levels of N nitrosamines. We conclude that the levels of nitrite typically found in water used for API manufacture are very low (<0.01 mg/L) and will not give rise to significant levels of N nitrosamines through reaction with basic secondary amines (pK_a > 9.5) in the majority of cases. The use of less basic amines, elevated processing temperatures, or low pH conditions in combination with elevated levels of nitrite have the potential to generate levels of N nitrosamines that could lead to significant quantities being present in the isolated API if the downstream processing does not provide an adequate purge. The kinetic models described may be used to risk assess specific situations or processes. For example, the addition of traces of dimethylamine to a nitrosation reaction is predicted to lead to the rapid, quantitative formation of N nitroso dimethylamine. Simple tertiary alkylamines can nitrosate via a dealkylative process, which is significantly slower than secondary amine nitrosation. Therefore, they do not represent a risk of N nitrosamine formation under conditions where there is no significant risk of secondary amine nitrosation.

KEYWORDS: *nitrosation, nitrite, N nitroso dimethylamine, nitrosamine, kinetic modeling*

INTRODUCTION

The contamination of members of the sartan family of drugs with low levels of potentially carcinogenic dialkyl N nitrosamines has been reported.[1−3] Additionally, more recent observations of dialkyl N nitrosamine contamination in non sartan medicinal products have led to regulatory requests that marketing authorization holders assess their portfolios for chemically synthesized active pharmaceutical ingredients (APIs) and associated drug products (DPs) that could be at risk of contamination with N nitrosamines.[4,5] This has led to a renewed interest in the understanding of potential risks of amine nitrosation during API manufacturing to ensure appropriate control and understanding of the formation of these species. While the stoichiometric use of sodium nitrite in a manufacturing process can represent a risk for nitrosamine formation if a secondary or tertiary amine is present, it is less clear whether small amounts of nitrite can lead to significant amounts of nitrosamine being formed (Scheme 1). Nitrite is a controlled impurity in water, with a WHO guideline limit of 3 mg/L[6] and a European limit of 0.5 mg/L.[7] The primary question this paper will focus on is: does trace nitrite, which may be present in water, constitute a significant risk in respect of the formation of nitrosamines during API manufacture? Due to the wide interest this question may attract, the outcome of the investigations will be discussed first, followed by the scientific basis of our conclusions, which includes discussions of the mechanisms of nitrosation in water and the kinetic modeling of several scenarios relevant to API manufacturing.

RISK OF NITROSATION BY TRACE NITRITE IN WATER

The nitrosation risk will clearly depend on the amount of nitrite present in process water. The WHO guideline limit for nitrite in potable water is not more than (NMT) 3 mg/L, based on safety considerations.[6] The minimum standard for water to be used in API manufacturing is potable water,[8,9] although higher pharmacopoeial grades (purified water and water for injection) are routinely used as well. The minimum standard for water to be used in drug product (DP) manufacturing is purified water.[8] Table 1 compares nitrite data for potable water for selected locations across the world where API manufacturing takes place. This data shows that potable water typically contains amounts of nitrite lower than the limit of quantitation (LOQ) of the analytical method used (frequently <0.012 mg/L), with the highest levels reported in the data collated here being 0.066 mg/L (or 66 ppb). The WHO guideline value of NMT 3 mg/L therefore represents a worst case scenario for the nitrite content of water used in API manufacturing, and a value of 0.01 mg/L may better represent the typical level of nitrite present in

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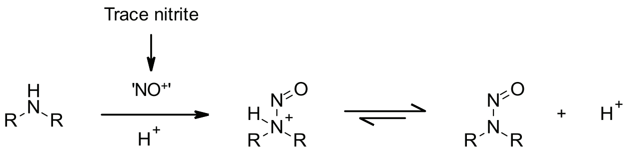
Scheme 1. Nitrosation of a Secondary Amine by Trace Nitrite



potable water. The nitrite level in purified water or water for injection does not have a specification, but since these grades of water are manufactured starting from potable water by methods that should further reduce nitrite levels, it can be assumed that the levels of nitrite will be substantially lower than 0.01 mg/L (for illustrative purposes, the level in purified water will be assumed to be <0.001 mg/L).

It is reasonable to assume that the worst case for nitrosamine formation from trace nitrite contained in process water is the quantitative reaction of the available nitrite with an amine to generate the corresponding nitrosamine. The amount of nitrite present will be directly proportional to the amount of water used in the process. A general simple equation (eq 1) therefore allows calculation of the absolute worst case amount of a nitrosamine impurity, relative to a substrate molecule, which could be generated from trace nitrite present in water

$$\begin{aligned}\text{ppb nitrosamine} &= \text{nitrite content (mg/L)}\\ &\quad\times \text{volume water (L/kg)} \times 10^3\\ &\quad\times \frac{\text{RMM nitrosamine}}{46}\end{aligned} \tag{1}$$

where ppb nitrosamine is the maximum amount of nitrosamine that could be formed relative to the substrate molecule, nitrite content is the measured nitrite level in water used, volume water is the the volume of water used during processing in L/kg of the substrate molecule (relative volume (RV)), RMM nitrosamine is the relative molecular mass of potential nitrosamine impurity, 46 is the RMM of nitrite.

To put the amounts potentially formed into perspective, the maximum quantity of *N* nitroso dimethylamine (NDMA) formed by a process using stoichiometric dimethylamine (DMA) has been calculated for a manufacturing process operation that uses 10 relative volumes (RV) of water (10 RV of water is 10 L of water/kg of the substrate). The results for process water containing varying levels of nitrite are compared to those in Table 2.

To understand the potential risk to patient safety, the maximum levels of NDMA that could be formed, shown in Table 2, need to be compared to the acceptable daily intake.[18]

Table 2. Absolute Worst Case Amount of NDMA Relative to the Reaction Product Formed from a Process Using a Stoichiometric Amount of DMA and 10 L/kg Water

nitrite level in water	NDMA formed in ppb
3 mg/L (WHO guideline limit)	48 300
0.01 mg/L (typical level in potable water)	161
0.001 mg/L (estimated level in purified water)	16

Table 3 shows the control limit for NDMA adjusted for less than lifetime exposure as per EMA Article 5(3) Q&A and the maximum daily dose,[18] expressed in ppb relative to the API. It can be seen that the worst case amount of NDMA potentially formed from typical maximum levels of nitrite in potable water (0.01 mg/L) will only be significant for products with a high daily dose and long treatment duration. Clearly, the quality of water used is crucial, as water containing nitrite at levels approaching the WHO guideline limit of 3 mg/L[6] will contain up to 300 times more nitrite than that found to typically be present in potable water. On the other hand, manufacturing plants using purified water are expected to have nitrite levels even lower than those observed in potable water (assumed here to be at 0.001 mg/L). Therefore, the nitrosation risk from trace nitrite in such waters will become vanishingly small, even assuming the worst case conversion of all of the available nitrite to the corresponding nitrosamine.

While the calculation of an absolute worst case amount of nitrosamine formation provides a useful context, a review of the literature has shown that the kinetics and mechanism of amine nitrosation in water have been extensively investigated. This understanding of the nitrosation of amines in water has enabled us to perform kinetic simulations, which provide an estimate of the likely level of conversion to the corresponding nitrosamine for a given reaction system (vide infra). The nitrosation of DMA to generate NDMA has been used as a model reaction under three scenarios that could arise during API manufacturing:

1. Trace sodium nitrite encountering trace quantities of dimethylamine to model the situation in a process operation using water containing trace nitrite downstream from the use of a secondary amine or where a secondary amine is present as a low level impurity (for example, in a tertiary amine reagent).
2. Trace sodium nitrite in the presence of high concen trations (1 M) of dimethylamine to model a situation where water containing traces of nitrite is used during the reaction or workup of a process using a secondary amine as a reactant or reagent.

Table 1. Sample Nitrite Levels in Potable Water

location	average nitrite (mg/L)	range (mg/L)	source
Coppell, Texas, USA	<0.023		City of Coppell[a]
Irvine, U.K.	<0.01	0 to <0.01	Scottish Water, Bradan D[b]
Macclesfield, U.K.	<0.012	0 to <0.012	United Utilities, Lamaload[c]
Montrose, U.K.	<0.01	0 to <0.01	Scottish Water, Lintrathen B[d]
Ringaskiddy, Ireland	<0.013	0 to <0.016	Irish Water, Cork harbor and city[e]
Sandwich, U.K.	0.0024	0 to 0.005	Southern Water, Beacon Hill[f]
Singapore	<0.03	0 to 0.066	PUB[g]
Stockholm, Sweden	<0.007	0 to <0.007	Stockholm Vatten Och Avfall[h]
Ulverston, U.K.	<0.0016	0 to <0.0016	United Utilities, Poaka Beck[i]

[a]Ref 10. [b]Ref 11 for KA11 5AP. [c]Ref 12. [d]Ref 11 for DD10 8EA. [e]Ref 13. [f]Ref 14. [g]Ref 15. [h]Ref 16. [i]Ref 17.

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Table 3. NDMA[a] Adjusted, Temporary Interim Control Limits Based on Treatment Duration and Daily Dose[b]

daily dose	NDMA adjusted, temporary interim control limit in ppb relative to the API			
	1 day to 1 month	1 month to 1 year	1 10 years	10 years to lifetime
10 mg	768 000	128 000	64 300	9600
100 mg	76 800	12 800	6430	960
1 g	7680	1280	643	96
10 g	768	128	64.3	9.6

[a]Ref 19. The temporary interim limits for NDMA and NMBA are 96 ng/day for lifetime exposure, while the limits for diethyl *N* nitrosamine (NDEA), *N* nitrosodiisopropylamine (DIPNA), and *N* nitrosoethylisopropylamine (EIPNA) are 26.5 ng/day. [b]Ref 18.

3. High concentrations (1 M) of sodium nitrite in the presence of traces of dimethylamine to model the presence of a secondary amine impurity during an aqueous nitrosation or diazotization reaction.[20]

The results from the simulations for the predicted quantities of NDMA that would be produced in 24 h, based on a 10 L/kg process using water containing various levels of nitrite at 25 °C are presented in Table 4 as a function of the process pH.[21] The conservative assumptions made in modeling the nitrosation of DMA and the fact that the model makes no allowance for the decomposition of nitrous acid[22−25] mean that the actual levels of NDMA formed should be lower than these predictions.

These simulations show that, as expected, the conversion of nitrite to the corresponding nitrosamine strongly depends upon the pH of the process and the concentrations of the reactants (nitrite and amine). Comparing the results shown in Table 4 with the NDMA adjusted interim control limits shown in Table 3 demonstrates that for amines similar to DMA (i.e., simple dialkylamines with pK_as greater than 9.5; a discussion of the effect of amine pK_a on the rate of nitrosation may be found later in this work):

- The risk of nitrosamine formation from low level amine impurities and trace nitrite in potable water (scenario 1 type processes) is negligible.
- The risk of nitrosamine formation from stoichiometric amines and trace nitrite in potable water (scenario 2 type processes) is negligible at the typically observed nitrite levels (<0.01 mg/L). Nitrosamine formation may become significant at lower pH values (<7) in situations where nitrite levels are approaching the WHO limit of 3 mg/L.
- In all situations where stoichiometric nitrite is used in the presence of even low level amine impurities (scenario 3 type processes), there is a significant risk of nitrosamine formation.

Note that these conclusions are not applicable to less basic alkylamines (pK_a ≤ 9.5) or processes run at elevated temperatures where rates of nitrosation are expected to be higher (vide infra).

Additionally, given the reported reduced rate of nitrosamine formation from tertiary amines compared to that from secondary amines;[26] basic tertiary amines with pK_as >9.5 are not expected to generate significant levels of nitrosamine by nitrosation, resulting from trace levels of nitrite in water.

Having laid out the context and the high level conclusions from our simulations, the following sections will review the mechanism and kinetics of the aqueous nitrosation of amines and explain how this knowledge may be applied to simulate the nitrosation of secondary amines. Naturally, this understanding may be used when assessing API production processes to determine the risk that the isolated API is contaminated with significant levels of *N* nitrosamines. In addition, the nitrosation of tertiary amines, the effects of amine pK_a and temperature on the rate of nitrosation, and nitrosating agent scavengers, which

Table 4. Summary of Predicted Quantities of NDMA Produced Based on a 10 Relative Volume Process as a Function of pH at 25 °C for 24 h with Added 1 M NaCl[a,b]

Scenario	DMA conc	Nitrite conc	ppb of NDMA produced at reaction pH			
			pH 3.15	pH 5	pH 7	pH 9
1a	1 mM[c]	0.01 mg/L[d]	3.6×10^{-3}	9.9×10^{-5}	1×10^{-6}	1×10^{-8}
1b	1 mM[c]	3 mg/L[e]	1.5	5.3×10^{-2}	5.3×10^{-4}	5.3×10^{-6}
2a	1 M	0.01 mg/L[e]	3.5	9.9×10^{-2}	1×10^{-3}	1×10^{-5}
2b	1 M	3 mg/L[d]	1450	53	0.53	5.3×10^{-3}
3	1 mM[c]	1 M	740000	740000	54400	560

[a]Key: risk of significant nitrosamine formation (NDMA) based on the lifetime exposure for a 1000 mg daily dose with an acceptable intake (AI) of 96 ng/day green = negligible risk, amber = level >30% of AI potential risk, red = level >AI significant risk. [b]Chloride is a known catalyst of nitrosation (vide infra) and its inclusion therefore provides a worst case rate of nitrosation while also mimicking the use of a brine wash or the use of HCl to acidify the $NaNO_2$ used in a nitrosation. [c]1 mM corresponds to 0.045% w/w relative to the limiting reagent based on a process dilution of 10 relative volumes (10 L/kg). [d]0.01 mg/L nitrite concentrate corresponds to the typical observed level for potable water and equates to a concentration of 2.2×10^{-7} M. [e]3 mg/L nitrite concentration corresponds to the WHO guideline limit for potable water and equates to a concentration of 6.5×10^{-5} M.

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are also relevant in the API risk assessment process, will all be discussed.

■ NITROSATION OF AMINES IN WATER

Historically, the chemistry of aniline nitrosation was studied extensively due to its importance in the manufacture of azo dyes, while the more recent discovery that dialkyl *N* nitrosamines are carcinogenic[27] has led to a greater focus on secondary amine nitrosation and *N* nitrosamine formation. The most widely used nitrosating agent is aqueous nitrous acid, which is a weak acid with a pK_a of 3.15 at 25 °C.[28] Due to its known instability with respect to decomposition to nitric oxide,[22−25] nitrous acid is usually generated in situ by the action of an acid on a nitrite salt. The reactions of nitrous acid and species derived from it with amines have been extensively studied, and the observed kinetics has been shown to vary with the acidity of the reaction solution and the concentration of nitrous acid. A thorough rationalization of these observations was provided by Ridd,[29] and wide ranging reviews of the field of nitrosation have been compiled by Turney and Wright[30] and Challis and Challis.[31] More recently, monographs by Williams[32,33] have provided significant coverage of the kinetics and mechanism of nitrosation. A more focused overview of the formation of *N* nitroso compounds with relevance to their potential formation in vivo was provided by Mirvish.[34] Additionally, their formation and presence in foods,[35] the environment,[36,37] and their formation in cosmetics and skincare products have been reviewed.[38,39] Nitrosation of amines may also occur in organic solvents with organic soluble nitrosating agents such as dinitrogen trioxide, dinitrogen tetroxide, nitrosyl chloride, and alkyl nitrites being effective reagents.[31] Nitrosation in organic solvents has been excluded from the scope of this review.

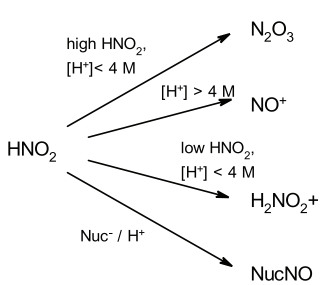
Aqueous Solutions of Nitrous Acid. Different nitrosating species can be generated in aqueous, acidic, solutions of nitrous acid depending upon the conditions (Scheme 2). These may act in parallel and are discussed in more detail below.[40]

In acidic solutions (<4 M H^+, pH > −0.6) containing higher concentrations of nitrous acid, the dimerization of nitrous acid to form dinitrogen trioxide (N_2O_3, the anhydride of nitrous acid) is a significant factor in the behavior of nitrous acid (Scheme 3). The equilibrium constant, $K_{N_2O_3}$ (eq 2), for this reaction has been reported as 3×10^{-3} M^{-1} at 25 °C[41]

$$K_{N_2O_3} = \frac{[N_2O_3]}{[HNO_2]^2} \tag{2}$$

Under more acidic conditions (>4 M H^+, pH <−0.6), nitrous acid is stoichiometrically converted to the nitrosonium ion (NO^+) (Scheme 4), which may be observed spectroscopically.[42]

Scheme 2. Summary of the Active Nitrosating Species That May Be Present in Aqueous Solutions of Nitrous Acid



Scheme 3. Dimerization of Nitrous Acid to Form N_2O_3

$$2\,HNO_2 \xrightleftharpoons{K_{N2O3}} N_2O_3 + H_2O$$

Scheme 4. Formation of the Nitrosonium Ion from Nitrous Acid

$$H_3O^+ + HNO_2 \rightleftharpoons NO^+ + 2\,H_2O$$

The strongly acidic conditions required to do this are unlikely to occur in many chemical processes, and this nitrosation pathway has therefore not been considered further.

Under conditions that are more relevant to the potential nitrosation by nitrite in process water, i.e., moderate acidities and low concentrations of nitrous acid, dinitrogen trioxide ceases to be the dominant nitrosating species. Under such conditions, an acid catalyzed process with a first order dependence upon the concentrations of nitrous acid, the substrate and H^+ is observed instead (eq 3). The observed nitrosation kinetics could arise from two kinetically indistinguishable mechanisms: one of which involves the nitrous acidium ion ($H_2NO_2^+$), while the second is via traces of NO^+ generated by the protonation of nitrous acid[43]

$$\text{rate} = k_{obs}[HNO_2][H^+][\text{substrate}] \tag{3}$$

The remaining significant species formed from HNO_2 in aqueous media are nitrosyl species formed by the reaction of nonbasic nucleophiles, such as halide ions,[44,45] thiocyanate,[46] thiourea, and thiosulfate. Such a reaction is exemplified in Scheme 5, wherein the added nucleophile, chloride, reacts with nitrous acid and H^+ to form nitrosyl chloride, a nitrosyl nucleophile adduct, which is an active nitrosating species. This scenario is likely to be common during API manufacturing processes, when, for example, salts are present during aqueous workup operations. The formation of nitrosyl chloride shown in Scheme 5 has been studied experimentally, and an equilibrium constant, K_{ClNO}, of 1.1×10^{-3} M^{-2} (eq 4) has been determined in water at 25 °C[44]

$$K_{ClNO} = \frac{[ClNO]}{[HNO_2][H^+][Cl^-]} \tag{4}$$

Kinetics of the Nitrosation of Secondary Amines in Water. *Nitrosation by Dinitrogen Trioxide (N_2O_3).* A wide range of amines have been used to study the kinetics of nitrosation by N_2O_3, including dimethylamine and diethyl amine.[47] At low amine concentrations, the formation of N_2O_3 from HNO_2 is normally fast relative to its consumption and a third order rate law is observed (eq 5). The observed rate constant, k_{obs}, in this rate law is a complex rate constant, which is composed of the second order rate constant for nitrosation of the amine by N_2O_3, $k_{N_2O_3}$, values of which have been reported,[47] and the equilibrium constant for the formation of N_2O_3, $K_{N_2O_3}$[41]

$$\text{rate} = k_{obs}[HNO_2]^2[R_2NH] \text{ where } k_{obs} = k_{N_2O_3} \times K_{N_2O_3} \tag{5}$$

In reality, the situation in water is slightly more complex as nitrous acid and the amine both undergo ionization dependent

Scheme 5. Formation of Nitrosyl Chloride from Nitrous Acid

$$HNO_2 + Cl^- + H^+ \rightleftharpoons ClNO + H_2O$$

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upon the reaction pH (Scheme 6). A modification to eq 5 to account for the fraction of the reactants in the desired form at the reaction pH is therefore required (eq 6).[48] Calculating the product of the fractions in the desired form as a function of pH makes it possible to generate a pH profile for the nitrosation of an amine (Figure 1). From this pH profile, which is proportional to the rate of reaction, it is clear that for an amine with a pK_a significantly greater than the pK_a of nitrous acid there is a clear optimum in the rate of nitrosation in the region of the pK_a of nitrous acid

$$\text{rate} = k_{N_2O_3}K_{N_2O_3}[HNO_2]_T^2 \cdot f_{HNO_2}^2 [R_2NH]_T \cdot f_{R_2NH}$$

where

$$f_{HNO_2} = \frac{[H^+]}{K_{a\,HNO_2} + [H^+]} \text{ and } f_{R_2NH} = \frac{K_{a\,R_2NH}}{K_{a\,R_2NH} + [H^+]} \qquad (6)$$

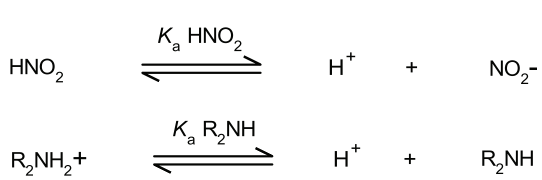
Nitrosation via the Nitrous Acidium Ion ($H_2NO_2^+$). The nitrosation reactions of a wide range of nucleophiles have been studied with dilute nitrous acid under mildly acidic conditions where the nitrous acidium ion is the dominant nitrosating species. These studies have found the rate law of eq 3 to apply with the reported rate constants for neutral nucleophiles tending toward a limit of 7000 M^{-2} s^{-1} at 25 °C, which has been taken as the diffusion controlled limit for this complex rate constant.[49]

While there are no reports of studies of secondary dialkylamine nitrosation under these conditions, it is reasonable to correct for the ionization of nitrous acid and the amine as a function of pH as above (see eq 6).[50] Using this approach gives an expression for the pH dependent behavior of the nitrosation of an amine by $H_2NO_2^+$ (eq 7). Plotting the behavior of the logarithm of the pH dependent terms in eq 7 as a function of pH (Figure 2) shows that the reaction rate will have little dependence upon the pH below the pK_a of nitrous acid (3.15) and that the rate will fall at high pH[51]

$$\text{rate} = k_{obs}[HNO_2]_T \cdot f_{HNO_2} [R_2NH]_T \cdot f_{R_2NH} [H^+] \qquad (7)$$

Nitrosation via Nitrosyl Halides (ClNO and BrNO). The kinetics of the reaction between dimethylamine and nitrosyl chloride and bromide have been studied.[52] Generally, the formation of the nitrosyl halide is fast and may be treated as a rapid pre equilibrium, which occurs prior to a rate limiting reaction between the nitrosyl halide and the amine. This results in a rate law with a first order dependence upon the concentrations of HNO_2, amine, halide, and H^+ (eq 8). Values of the equilibrium constant K_{XNO} (Scheme 5, eq 4 X = Cl or Br) have been determined for nitrosyl chloride and nitrosyl bromide (Table 5). These show the nitrosyl bromide formation to be more favorable than the nitrosyl chloride formation and suggest that the level of nitrosyl halide formation will increase with temperature. The second order rate constants, k_{BrNO}, for the reaction between nitrosyl bromide and a range of secondary amines show little sensitivity to the pK_a of the amine or its

Scheme 6



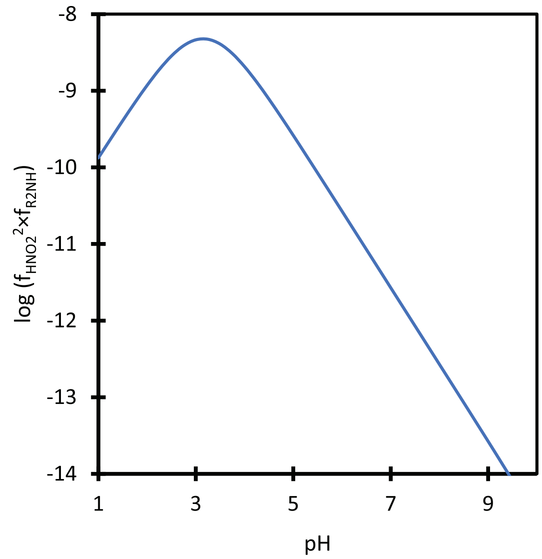


Figure 1. pH dependence of the rate law for the nitrosation of dimethylamine by N_2O_3.

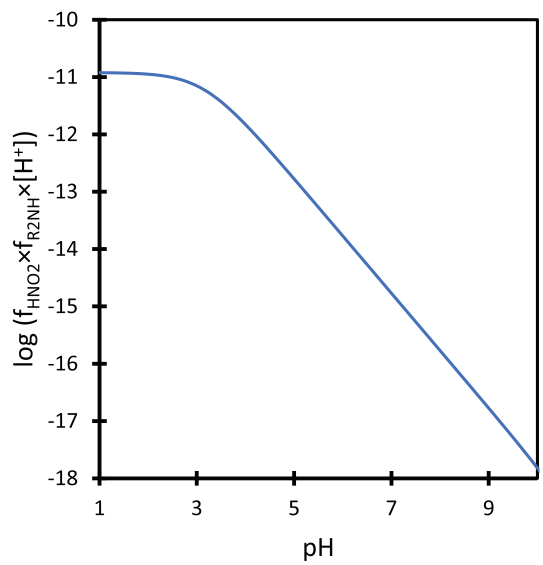


Figure 2. pH dependence of the rate law for the nitrosation of dimethylamine by $H_2NO_2^+$.

Table 5. Equilibrium Constants, K_{XNO} (M^{-2}), for Nitrosyl Bromide[45] and Chloride[44] Formation in Water

X	0 °C	25 °C
Br	2.2×10^{2}	5.1×10^{2}
Cl	5.6×10^{4}	1.1×10^{3}

structure. Rate constants of (0.7×10^7)–(6.7×10^7) M^{-1} s^{-1} at 25 °C were obtained for a pK_a range of 8–11.25.[53] This insensitivity of the rate constant to structural and electronic features has been taken as evidence that the reaction occurs at close to limit for an encounter controlled process.[54] Nitrosyl fluoride has also been isolated,[55] although its kinetics and equilibria have not been studied as thoroughly as nitrosyl chloride and bromide. There is evidence for catalysis by iodide in the diazotization of aniline;[56] however, nitrosyl iodide has not been isolated

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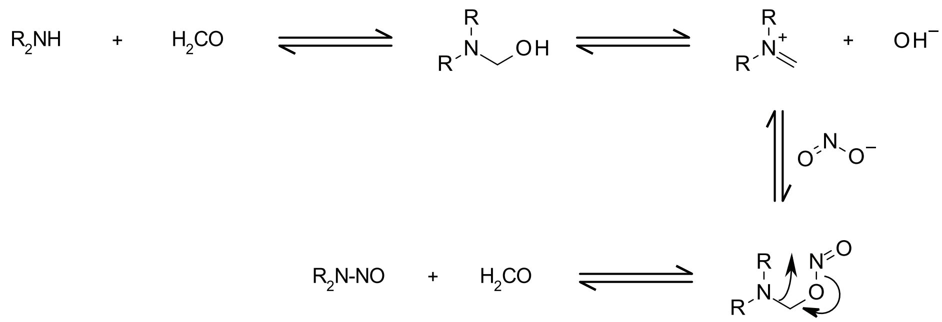
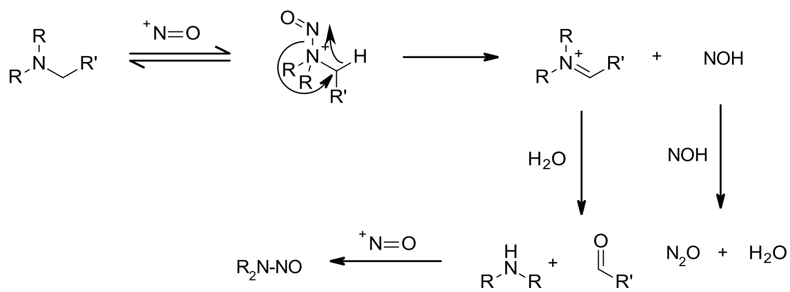
Scheme 7. Mechanistic Scheme for the Formaldehyde Catalyzed Nitrosation of a Secondary Amine by Nitrite



Scheme 8. Proposed Mechanistic Scheme for the Nitrosation of Simple Tertiary Amines



$$\text{rate} = k_{XNO}K_{XNO}[HNO_2]_T \cdot f_{HNO_2}[R_2NH]_T \cdot f_{R_2NH}[X^-][H^+] \quad (8)$$

The pH dependent terms in the rate law for nitrosation by nitrosyl halides (eq 8) are the same as those in eq 7. This means that the pH rate behavior of the reaction between nitrosyl halides and dialkylamines will follow the form of Figure 2.

Nitrosation by Nitrite. Nitrite is not normally a nitrosating agent under neutral to basic conditions. However, it has been observed that nitrosation of secondary amines can occur in the presence of formaldehyde, chloral, and benzaldehyde at neutral to basic pH.[57,58] Kinetic studies of this reaction[59,60] have confirmed the proposal that the reaction proceeds via the formation of an iminium ion intermediate through the reaction of formaldehyde with a secondary amine.[57] This reactive intermediate then reacts with nitrite to form the nitrosamine coupled with the elimination of formaldehyde (Scheme 7). The observed kinetics are complex and dependent upon the nature of the amine as this affects the level of the intermediate carbaminol that is formed in the initial rapid pre equilibrium. These equilibria have been studied for a limited range of dialkylamines. Electronic structure calculations have shown that the energetics of the reaction is sensitive to alkyl substitution on the aldehyde catalyst,[61] rationalizing the fact that catalysis was not observed in the case of acetone.[57]

The kinetic studies employed an initial rate based method ology and showed the initial rate of the formaldehyde catalyzed nitrosation of DMA to fall as the pH was increased from 3.6 to 9. As the pH approached 9, the change in the initial rate was observed to slow and tend toward a constant value.[59] Comparison of the observed initial rate of 1×10^{-6} M s^{-1} at pH 9 and 37 °C for a reaction between formaldehyde (1 M), dimethylamine (1 M), and sodium nitrite (0.053 M) with the predicted initial rate using the approach outlined later for the reaction based on N_2O_3 and $H_2NO_2^+$ at 25 °C of 2.6×10^{-11} M s^{-1} clearly demonstrates the catalytic effect of formaldehyde at high pH. At lower pH (3−5) and typical concentrations of nitrite (0.01−0.07 M), the formaldehyde catalyzed pathway was competitive with nitrosation via N_2O_3 and $H_2NO_2^+$.

The presence of stoichiometric levels of formaldehyde, or other catalytically active aldehydes, in systems containing nitrite under neutral to basic conditions is therefore a risk with respect to dialkyl *N* nitrosamine formation if dialkylamines are present.

Kinetics and Mechanism of the Nitrosation of Tertiary Alkylamines in Water. Despite the fact that many textbooks state that tertiary amines do not react with nitrous acid, they in fact do in a dealkylative process that yields a dialkyl *N* nitrosamine product.[62,63] Further study has shown the products of the reaction to be a dialkyl *N* nitrosamine, an aldehyde, and nitrous oxide (N_2O).[64] Mechanistically, it is proposed that the tertiary amine undergoes reversible nitrosation to form an *N* nitroso ammonium ion, which slowly eliminates nitroxyl (NOH) to form an iminium ion. Rapid hydrolysis and nitrosation follow to give the observed products (Scheme 8). Nitroxyl dimerizes to give nitrous oxide, the observed byproduct.[65] A subset of complex tertiary amines have been shown to undergo facile nitrosative dealkylation to yield *N* nitrosamines.[66]

The kinetics of the nitrosative dealkylation of triethylamine has been studied by an initial rate methodology in buffered aqueous acetic acid at 74.8 °C and pH > 3.1.[26] The kinetically determined rate law (eq 9) is consistent with a rapid reversible nitrosation of triethylamine followed by a slow rate limiting elimination of nitroxyl to form an iminium ion intermediate (Scheme 8). In aqueous media, this rapidly hydrolyzes to give acetaldehyde and diethylamine, which rapidly undergoes nitrosation to form diethyl *N* nitrosamine (NDEA)

$$\text{rate} = k[HNO_2][R_3N][H^+] \quad (9)$$

A number of drugs and complex molecules containing tertiary amine functionality have been identified that exhibit atypically high rates of nitrosative dealkylation relative to the rates seen for simple tertiary alkylamines.[66] A clear example of this behavior

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has been reported for aminopyrine **1**,[67,68] which rapidly liberates NDMA upon treatment with nitrous acid, and is believed to react through its enamine function to give an intermediate that directly eliminates NDMA (Scheme 9). Gramine **2**,[69] a component of barley malt, and related 2 dimethylaminomethylpyrrole[70] **3** both exhibit rapid nitrosative dealkylation and give high yields of NDMA. In both cases, it is postulated that the electron rich heterocycle participates in the elimination of NDMA from an initially formed *N* nitroso ammonium ion (Scheme 10). While relatively rare, such structural features that may impact on the reactivity of tertiary amines should be considered in complex structures that contain tertiary amines.

Mechanism of the Nitrosation of Quaternary Ammonium Ions in Water. The nitrosation of a limited number of quaternary ammonium ions has been reported under forcing conditions.[71] It was proposed that nitrosation was preceded by a demethylation reaction to generate a tertiary amine (Scheme 11), which then underwent nitrosative dealkylation followed by nitrosation (Scheme 8). Dealkylation of quaternary ammonium ions is a known issue in their use as phase transfer catalysts[72] and principally occurs via either Hoffman elimination in the case of alkyl groups possessing a β hydrogen atom[73] or nucleophilic substitution for quaternary ammonium ions possessing methyl[74] or benzyl groups.[75]

No detailed kinetic studies of the nitrosation of quaternary ammonium ions have been reported. However, based on the known potential for quaternary ammonium ions to dealkylate under appropriate conditions and the published kinetics and mechanism of tertiary amine nitrosation, the proposed mechanism is reasonable. It therefore follows that quaternary ammonium ions do not directly undergo nitrosation. Additionally, any generation of nitrosamines from quaternary ammonium ions under nitrosating conditions will be slower than the nitrosative dealkylation of tertiary amines that they must first be converted into.

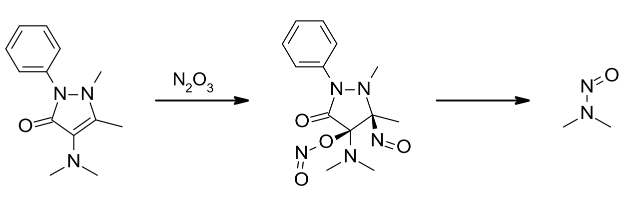
Nitrous Acid Scavengers. Due to the instability of solutions of nitrous acid,[22−25] it is normally used in excess. Nitrosation and diazotization reactions therefore usually contain nitrosating species at the end of the reaction that can carry forward with the reaction product and present a risk of nitrosamine formation in the downstream chemistry if they encounter a secondary amine. The use of a nitrous acid scavenger as part of the workup following a nitrosation reaction has the potential to mitigate this risk by reducing the amount of nitrosating potential that carries forwards with the product.

A range of compounds have been investigated as nitrous acid scavengers in vivo to prevent dialkyl nitrosamine formation in the stomach by dietary nitrite.[76] The inhibition of nitrosation of secondary amines has also been considered from the point of view of cosmetic products.[38] Inhibition of nitrosation has been brought about by the addition of materials that react more rapidly, with the active nitrosating species, than any secondary amines that are present. The generation of stable non nitrosating products from the nitrosating agents in this manner suppresses the formation of dialkyl *N* nitrosamines. Broadly speaking, the inhibitors fall into one of two classes; either they react with nitrous acid via nitrosation to give stable products or they act as reducing agents to reduce nitrous acid to various nitrogen oxides.[24]

The relative reactivities of a range of nitrous acid scavengers that work by converting the nitrosating agent into stable non nitrosating products have been compared by Williams.[77] These include azide (hydrazoic acid),[78] urea,[79] sulfamic acid,[77,80] and hydroxylamine (Scheme 12).[81] Urea is relatively ineffective at high pH but has found considerable utility as a scavenger for the traces of nitrosating species that often contaminate concentrated nitric acid and can lead to nitrosated products being formed in addition to the desired nitrated products in nitrations.[77] The low pK_a of sulfamic acid means that it is in the reactive anionic form above pH 2[80] and it is commonly used as a scavenger of nitrous acid in kinetic experiments. Azide, hydroxylamine, hydrazine, and ammonia are all used as synthetic reagents and may help to remove nitrosating impurities when used in a synthesis, although it is unlikely that they will be used solely for this purpose. The reaction between azide and nitrous acid is so effective that nitrous acid is often used to destroy residual azide and thus control the explosivity risks associated with the potential for hydrazoic acid generation in processes using azides. In fact, it was the use of nitrous acid to destroy azide that is believed to have led to the contamination of valsartan with NDMA.[1]

Mixed Aqueous-Organic Solvent Systems. The preceding section has dealt with the nitrosation of amines in water. In the situations encountered during processing, it is likely that quantities of water miscible organic solvents may be present. The impact that these may have upon the nitrosation of amines has therefore been considered. Low levels of water miscible solvents (<20% w/w) are expected to have a relatively small effect based on the relative insensitivity of dissociation constants to low levels of a cosolvent for a wide range of water miscible solvents.[82] However, the presence of significant quantities (>50% w/w) of a second solvent is expected to change the constants used in the models in the following section due to the effects of solvation upon the rates of the reactions and position of the equilibria involved. Applying the quantitative predictions and predictive methodologies contained in this work to assess the likely behavior in systems where water is not the dominant solvent should be undertaken with extreme caution.

Scheme 9. Nitrosation of Aminopyrine to Liberate NDMA



■ DETAILED DISCUSSION AND RESULTS

The published studies of the kinetics and mechanism of amine nitrosation by nitrous acid, and species derived from it, provide a suitable basis upon which to model their formation under some conditions that are relevant to API manufacturing. As discussed earlier, the nitrosation of dimethylamine (DMA) to generate dimethyl *N* nitrosamine (NDMA) was used as a model reaction in three scenarios that could arise during API processing. Scenarios 1 and 2 address the risk posed by nitrite present in process water encountering secondary amines. Scenario 3 addresses the situation believed to have led to the contamination of valsartan with NDMA[4] and considers the nitrosation of DMA under the conditions of a synthetic nitrosation.

Scenario 1. Trace sodium nitrite encountering trace quantities of dimethylamine to mimic the situation that

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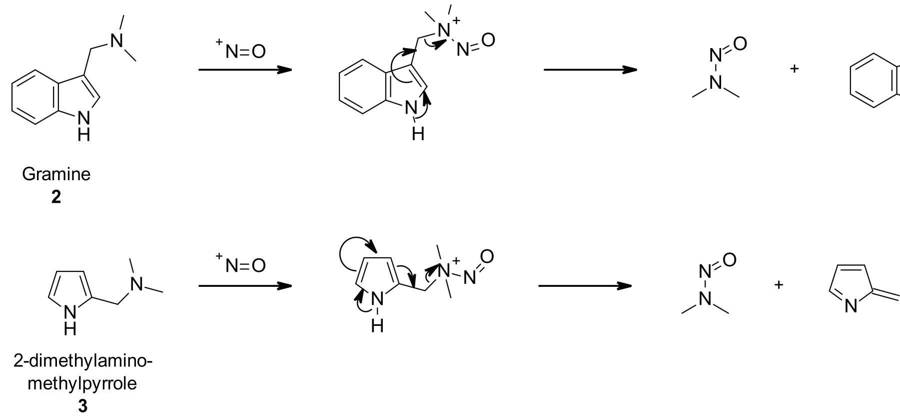
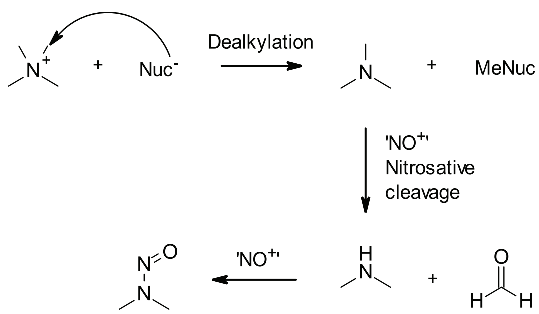
Scheme 10. Proposed Eliminative Mechanism for the Formation of NDMA from Gramine 2 and 2 Dimethylaminomethylpyrrole 3



Gramine
2

2-dimethylamino-
methylpyrrole
3

Scheme 11. Proposed Route of Formation of NDMA from Tetramethyl Ammonium Chloride



could arise in a process operation using water containing trace nitrite downstream from the use of a secondary amine or where a secondary amine is present as a low level impurity (for example, in a tertiary amine reagent).

Scenario 2. Trace sodium nitrite in the presence of high concentrations (1 M) of dimethylamine to model a situation where water containing traces of nitrite is used during the reaction or workup of a process using a secondary amine as a reactant or reagent.

Scenario 3. High concentrations (1 M) of sodium nitrite in the presence of traces of dimethylamine to model the presence of a secondary amine impurity during an aqueous nitrosation or diazotization reaction.

Two concentrations of nitrite in water will be used for the calculations as described in the high level discussion section and Table 1: the typical potable water level of 0.01 mg/L (or 2.2×10^{-7} M) and the WHO guideline limit of 3 mg/L (or 6.5×10^{-5} M),[6] which represents the worst case for potable water. The nitrite levels in purified water or water for injection do not have a specification, but since these grades of water are manufactured starting from potable water, it can be assumed that the levels of nitrite will be substantially lower than 0.01 mg/L (for illustrative purposes, the levels for purified water will be assumed to be 0.001 mg/L).

Using eqs 6–8, it is possible to estimate the initial rate of NDMA formation as a function of pH for a given concentration of DMA and nitrite if the rate and equilibrium constants are known. For nitrosation by dinitrogen trioxide (N_2O_3), a rate constant of 1.2×10^8 M^{-1} s^{-1} has been reported for the nitrosation of DMA at 25 °C,[47] while a rate constant of 3.1×10^7 M^{-1} s^{-1} is available for the nitrosation of DMA by nitrosyl chloride (ClNO) at 25 °C.[52] These were used in eqs 6 and 8 with the reported equilibrium constants for the formation of N_2O_3 (3×10^{-3} M^{-1})[41] and ClNO (1.1×10^{-3} M^{-2})[44] and pK_as of 3.15 for nitrous acid[28] and 10.91 for DMA.[83,84] A rate constant for the nitrosation of DMA by the nitrous acidium ion ($H_2NO_2^+$) is not known but the reported values for neutral nucleophiles tend toward a limiting value of 7000 M^{-2} s^{-1} at 25 °C.[49] This model is summarized in Scheme 13. Overall, this approach should overestimate the rate of NDMA formation at low concentrations of nitrite because it overestimates the rate of nitrosation via the nitrous acidium ion ($H_2NO_2^+$, eq 7) and makes no allowance for the decomposition of nitrous acid to nitric oxide (Scheme 14).[22−25]

Scenario 1: Trace Nitrite with Trace Amine. Initial rates of DMA nitrosation were calculated as a function of pH for the

Scheme 12. Nitrous Acid Scavenging Reactions of Hydrazoic Acid, Urea, Sulfamic Acid, and Hydroxylamine

$$HNO_2 + HN_3 \longrightarrow N_3NO + H_2O \longrightarrow N_2 + N_2O$$

$$2\,HNO_2 + CO(NH_2)_2 \longrightarrow 2\,N_2 + 3\,H_2O + CO_2$$

$$HNO_2 + NH_2SO_3H \longrightarrow N_2 + H_2O + H_2SO_4$$

$$HNO_2 + NH_2OH \longrightarrow N_2O + 2\,H_2O$$

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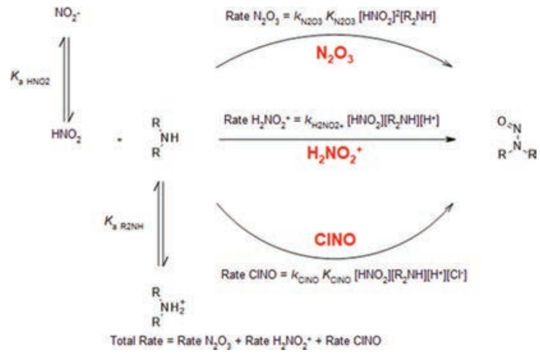
Scheme 13. Scheme Showing the Reactions Used to Simulate the Formation of Nitrosamines from Secondary Amines in an Aqueous Environment



Scheme 14. Decomposition of Nitrous Acid to Nitric Oxide

$$3\ HNO_2 \longrightarrow 2\ NO + HNO_3 + H_2O$$

reaction between trace nitrite (0.01 mg/L, 2.2×10^{-7} M) and trace amounts of secondary amine (1 mM)[85] in the presence of trace (1 mM) and stoichiometric chloride (1 M). Initial rate profiles versus pH were generated from these calculations (Figure 3a,b), which show the contributions of each of the nitrosating species in addition to the overall total rate. Both scenarios show a pH independent reaction at pH values below the pK_a of nitrous acid. Comparison of Figure 3a,b illustrates the potential for catalysis by chloride as the dominant nitrosating species switches from being $H_2NO_2^+$ at low chloride to ClNO with higher levels of chloride. Naturally, this has implications for processes that use a brine wash, especially if it is acidic. Substitution of the available constants for the formation[45] and reaction of BrNO[53] makes it possible to consider situations that generate bromide as a stoichiometric byproduct. The circles showing the total initial rate for all pathways will closely follow the line for one of the pathways when that pathway is strongly dominant.

While the initial rate calculation provides a clear view on the relative reactivities of the different nitrosating species, it does not indicate how long it will take for DMA to be converted into NDMA. This is an important consideration when process exposure times are relatively brief and was addressed using a kinetic simulation based on the rate laws for the three principal nitrosation pathways (eqs 6−8). Simulation of the nitrosation of DMA (1 mM)[85] by trace levels of nitrous acid (2.2×10^{-7} M) in the presence of chloride (1 M) shows the reaction to be extremely slow (see Figure 4), with a final conversion of 2.5% at the end of the simulation (1×10^8 seconds, ~3.2 years) based on the initial nitrous acid concentration. The predicted conversion over 24 h at 25 °C and pH 3.15 is very low with 0.002% of the available nitrous acid undergoing reaction to form NDMA. Performing a simulation with the worst case level of nitrous acid (6.5×10^{-5} M) gives, as expected, a faster reaction (see Figure S1)[86] with a final conversion of 3.5% at the end of the simulation (1×10^8 s, ~3.2 years) and a conversion of 0.003% after 24 h. While the conversions for the different water qualities appear to be similar, the amount of nitrosamine formed is not, as it also depends on the amount of available nitrite, which is the limiting reagent (see Table 4).

Scenario 2: Trace Nitrite with Stoichiometric Amounts of Amine. Using the approach taken previously (see above), the initial rates of nitrosation of DMA (1 M) by trace nitrite (2.2×10^{-7} M) were calculated as a function of pH at low (1 mM) and high levels of chloride (1 M). A comparison of the resultant pH initial rate profile for the reaction with that for a reaction involving trace DMA (1 mM) at high (1 M) chloride is given in Figure 5.[87] The reaction is, as expected, significantly faster as the 1000 times higher concentration of DMA leads to a 1000 fold increase in the rate of nitrosation due to the first order dependence upon the DMA concentration in eqs 6−8. The nitrous acid concentration is unchanged from the previous scenario, and therefore, the pH dependence and dominant nitrosating species are unchanged as may be seen if Figure 3b is compared to Figure 5.

A kinetic simulation of the nitrosation of dimethylamine (1 M) by trace nitrous acid (2.2×10^{-7} M) in the presence of chloride (1 M) was carried out (Figure 6). The reaction is faster

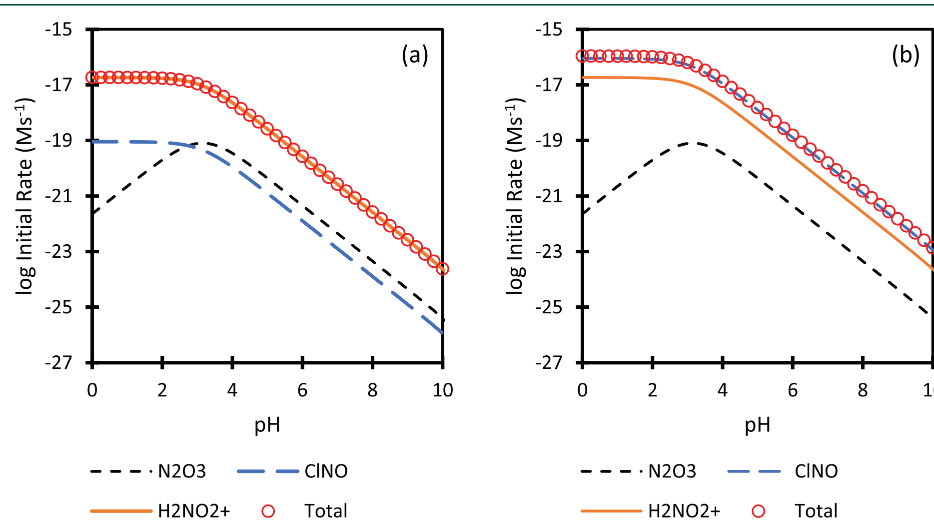


Figure 3. Simulated pH initial rate profiles showing the contributions of the principal nitrosating species for the nitrosation of trace DMA by water containing trace nitrite with either (a) trace chloride and (b) high chloride at 25 °C. (Conditions $[NO_2^-] = 2.2 \times 10^{-7}$ M, = [DMA] = 1 mM, and (a) $[Cl^-]$ = 1 mM or (b) $[Cl^-]$ = 1 M.)

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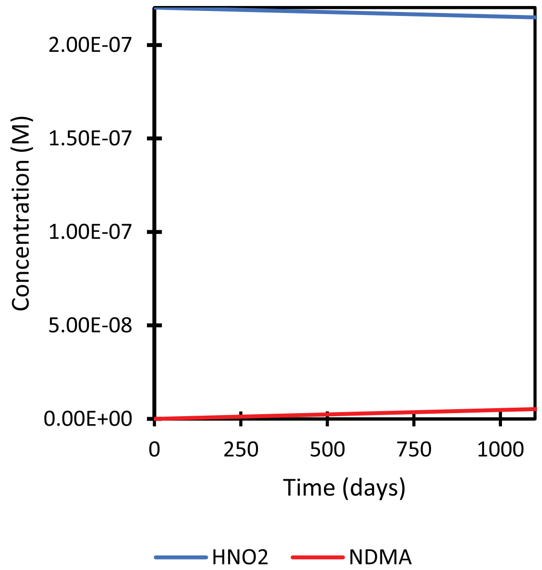

Figure 4. Simulated concentration profile for the aqueous nitrosation of DMA (1 mM) by $NaNO_2$ (2.2×10^{-7} M) in the presence of NaCl (1 M) at pH 3.15 and 25 °C.

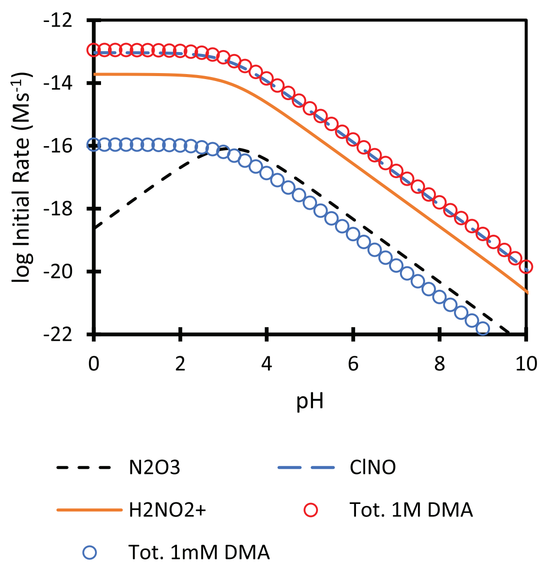

Figure 5. Simulated pH initial rate profiles showing the contributions of the principal nitrosating species for the nitrosation of DMA (1 M) in NaCl by traces of nitrite at 25 °C compared to the initial rate of trace DMA (1 mM). (Conditions $[NO_2^-] = 2.2 \times 10^{-7}$ M, [DMA] = 1 M, $[Cl^-]$ = 1 M.)

than with trace dimethylamine (1 mM, Figure 4),[88] with 580 days (5×10^7 seconds) being required to achieve complete conversion of the available nitrous acid to NDMA. Application of a more realistic upper limit for a reaction time of 24 h gave a significantly lower conversion of 2.1% based on the available nitrous acid. Using water that just meets the WHO guideline limit for nitrite of NMT 3 mg/L (6.5×10^{-5} M)[6] naturally gives a faster rate of reaction with a higher conversion of 2.9% occurring over 24 h. The 300 fold difference in the initial levels of nitrous acid used in these simulations means that they give rise to very different levels of NDMA (see Table 4).

Further simulations of the nitrosation reaction at pH values 5 and 9 show the rate of NDMA formation to decrease

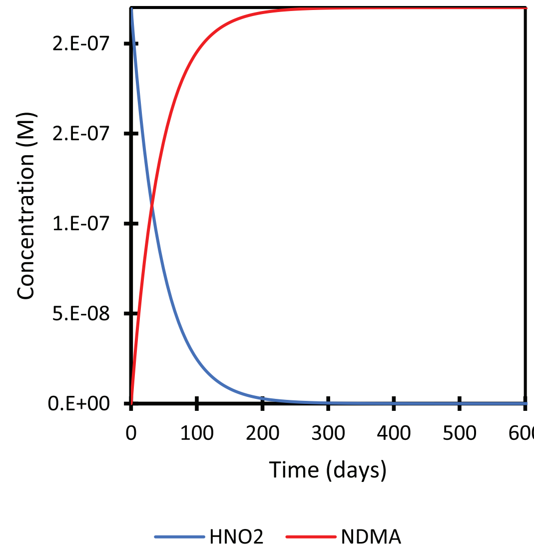

Figure 6. Simulated concentration profile for the aqueous nitrosation of DMA (1 M) by $NaNO_2$ (2.2×10^{-7} M) in the presence of NaCl (1 M) at pH 3.15 and 25 °C.

significantly as the pH is increased (Figure 7). This decrease in rate with increasing pH arises from the decreases in the concentrations of nitrous acid and H^+ as the pH is increased, which outweigh the increase in the concentration of unprotonated DMA. A comparison of the levels of NDMA predicted to be formed based on the different simulations undertaken is given in Table 4. The practical significance of the wide variation in the rate with pH is discussed more fully below.

Scenario 3: Stoichiometric Amounts of Nitrite. The calculation of the initial rates of DMA nitrosation as a function of pH for the reaction between a 1 M solution of sodium nitrite and traces of DMA (1 mM)[85] in the presence of trace (1 mM) and stoichiometric chlorides (1 M) gives a rather different picture (Figure 8) to that obtained for trace nitrite (Figures 3 and 5). From the resulting pH initial rate profiles, it is clear that under

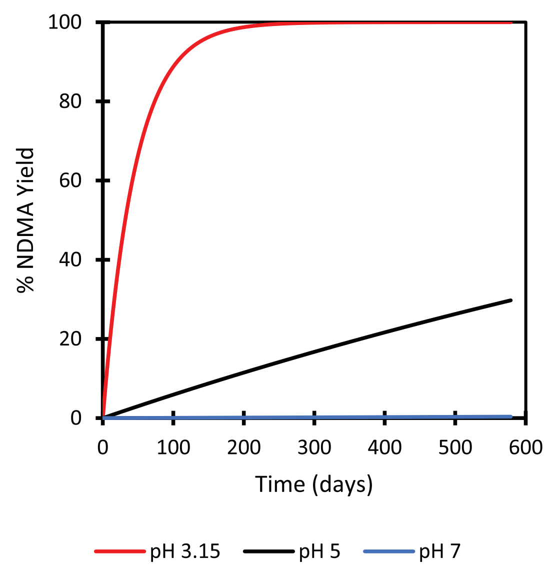

Figure 7. Simulated NDMA formation profiles as a function of pH for the aqueous nitrosation of DMA (1 M) by trace nitrite (2.2×10^{-7} M) in aqueous NaCl (1 M) at 25 °C.

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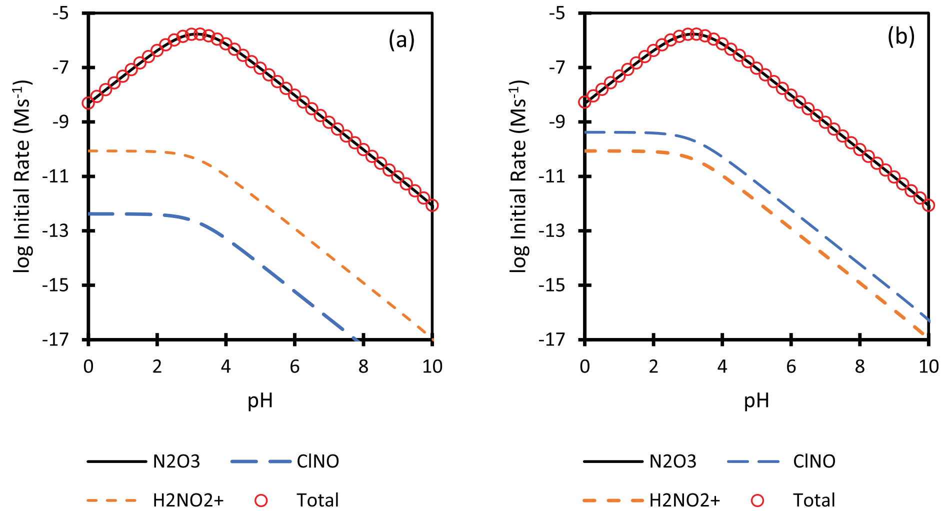

Figure 8. Simulated pH initial rate profiles showing the contributions of the principal nitrosating species for the nitrosation of trace DMA under nitrosating conditions with either (a) trace chloride or (b) high chloride at 25 °C. (Conditions $[NO_2^-]$ = 1.0 M, [DMA] = 1 mM, and (a) $[Cl^-]$ = 1 mM or (b) $[Cl^-]$ = 1 M.)

these conditions the rate shows a clear maximum in the region of the pK_a of nitrous acid and that the dominant pathway is nitrosation by N_2O_3. Comparison of Figure 8a,b shows that, while the rate of nitrosation via nitrosyl chloride is accelerated significantly by the presence of high levels of chloride, it is still uncompetitive relative to the rate of nitrosation by N_2O_3 at such high concentrations of nitrous acid.

The second order dependence upon the nitrous acid concentration in the rate law for the reaction via N_2O_3 (eq 6) means that the rate of reaction drops rapidly as the nitrous acid concentration is reduced. This is illustrated in Figure 9, which shows the effect of varying the nitrous acid concentration upon the initial rate of nitrosation of DMA (1 M) in the NaCl solution (1 M) at 25 °C and pH 3.15. A nonlinear decrease in the initial rate is seen as the nitrite concentration is reduced due to the dominant contribution of N_2O_3 to the rate at nitrite concentrations above 5×10^{-5} M.

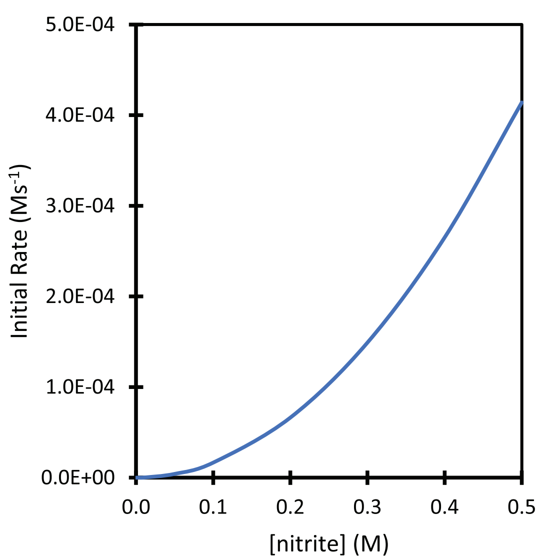

Figure 9. Initial rate of DMA (1 M) nitrosation in aqueous NaCl (1 M) as a function of sodium nitrite concentration at pH 3.15 and 25 °C.

Simulation of the nitrosation of DMA (1 mM)[85] by sodium nitrite (1 M) in the presence of chloride (1 mM) at pH 3.15 and 25 °C found the reaction to be fast (Figure 10) with a half time of less than 10 min. It is therefore reasonable to conclude that any traces of secondary amine that encounter nitrosating conditions will be stoichiometrically converted to the corresponding dialkyl *N* nitrosamine. Simulations of the nitrosation reaction at pH values 5, 7, and 9 show the rate of NDMA formation to decrease significantly as the pH is increased (Figure 11).

The pH initial rate profiles for nitrosation in aqueous NaCl (1 M) are compared for the three scenarios in Figure 12, which shows that scenario 3 is significantly faster than scenarios 1 and 2. Above the optimum nitrosation pH (approximately 3.15), the initial rate predicted for scenario 3 is over 10^{10} times faster than that expected for the reaction of trace nitrite with trace DMA (scenario 1). The initial rates of nitrosation expected for water

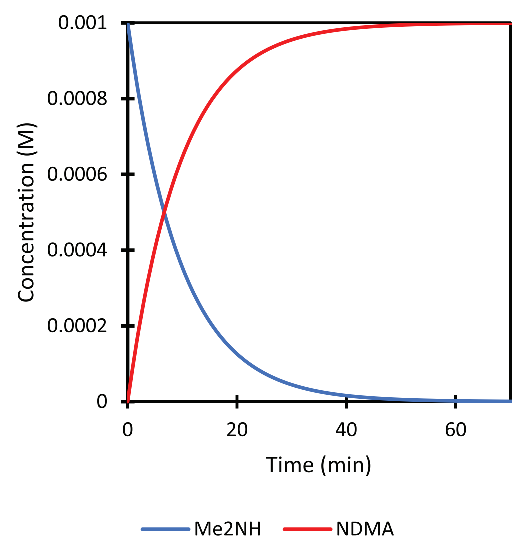

Figure 10. Simulated concentration profile for the aqueous nitrosation of DMA (1 mM) by $NaNO_2$ (1 M) in the presence of NaCl (1 mM) at pH 3.15 and 25 °C.

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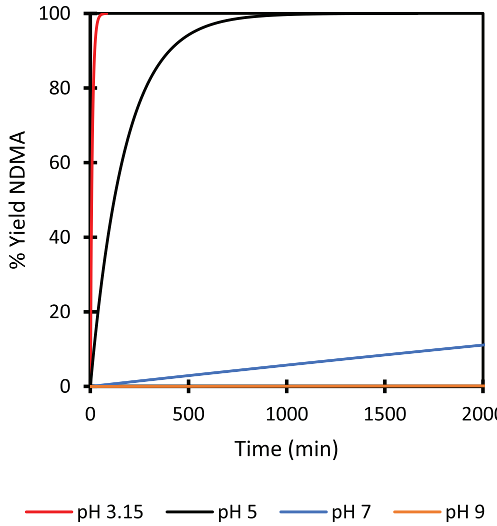

Figure 11. Simulated NDMA formation profiles as a function of pH for the aqueous nitrosation of DMA (1 mM) by $NaNO_2$ (1 M) in the presence of NaCl (1 mM) and 25 °C.

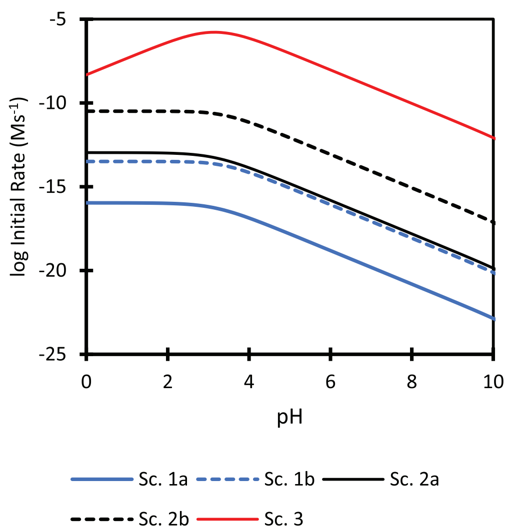

Figure 12. Comparison of the predicted initial rate pH profiles in aqueous NaCl (1 M) at 25 °C for scenarios 1 3. (Conditions: Sc. 1a $[HNO_2] = 2.2 \times 10^{-7}$ M and [DMA] = 1 mM, Sc. 1b $[HNO_2] = 6.5 \times 10^{-5}$ M and [DMA] = 1 mM, Sc. 2a $[HNO_2] = 2.2 \times 10^{-7}$ M and [DMA] = 1 M, Sc. 2b $[HNO_2] = 6.5 \times 10^{-5}$ M and [DMA] = 1 M, Sc. 3 $[HNO_2]$ = 1 M and [DMA] = 1 mM.)

that just meets the WHO guideline for nitrite in potable water of NMT 3 mg/L (6.5×10^{-5} M)[6] are also shown as dashed lines for scenarios 1 and 2. These reactions are over 300 times faster than the predictions for reactions using potable water with a more typical nitrite level (0.01 mg/mL, 2.2×10^{-7} M), which largely reflects the 300 fold increase in nitrite concentration.

The simulations undertaken have used numerical integration to solve the differential equations (rate laws) that describe the reaction through the three different pathways to generate concentration versus time profiles for DMA, nitrite, and NDMA. This approach correctly deals with the change in the reactant concentrations over the course of the simulation. However, at low degrees of conversion, the change in the reactant concentrations due to the reaction is less significant and a zero order approximation may be applied to calculate the concentration of NDMA formed based on the initial rate of reaction (eq 10).[89] Doing this for scenario 1 (2.2×10^{-7} M HNO_2, 1 mM DMA in 1 M NaCl) gives an NDMA concentration of 4.8×10^{-12} M after 24 h, which equates to a conversion of 0.0022% based on the available nitrite, in agreement with the above simulation results. While not as accurate as the kinetic simulation methodology used in the simulations described herein, the zero order approximation has the advantage that it can be readily applied without reference to specialist software. The assumptions implicit in the zero order approximation mean that eq 10 will always overestimate the degree of reaction relative to a full kinetic simulation. It is therefore reasonable to use eq 10 as an alternative to a full kinetic simulation, which it will most closely follow at low degrees of reaction. An effective rate of nitrosation calculator based on eq 10 can be implemented in a spreadsheet package

$$[\mathrm{NDMA}] = (\mathrm{init.\ rate}_{N_2O_3} + \mathrm{init.\ rate}_{H_2NO_2^+} + \mathrm{init.\ rate}_{ClNO}) \times \text{time in seconds} \quad (10)$$

The percentage conversions of nitrite or DMA to NDMA or the concentrations of NDMA predicted for the three scenarios are not readily comparable to the current interim acceptable intakes for dialkyl *N* nitrosamines, which are set in terms of ng/day.[19] For example, the temporary interim control limit for NDMA is 96 ng/day for lifetime exposure. The levels of NDMA generated in the simulations were therefore used to calculate an absolute amount of NDMA to facilitate comparison. To do this, a relatively dilute reaction that used 10 L of water/kg of the intermediate being processed (10 relative volumes) was considered, making it possible to calculate a mass of NDMA generated. Assuming the NDMA generated is not purged[90] and that 1 kg of the intermediate used to scale the calculation gives 1 kg of API it is possible to express the level of NDMA predicted to be formed in ppb versus the API.[21] The resultant values are shown and compared in Table 4. A simple comparison of these levels to the temporary interim control limit for NDMA of 96 ng/day for lifetime exposure would be appropriate for a dose of 1 g/day and lifetime exposure. However, this is unlikely to always be the case, and Table 3 shows the adjusted interim control limit for NDMA, expressed in ppb, based on less than lifetime exposure as per ICH M7,[91] the recent EMA guidance,[18] and the daily dose.

Extension to Other Secondary Amines. The use of DMA as the model amine in the simulations raises two potential issues: first, any sensitivity of the absolute reactivity of amines to their structure, and second, the variation of the fraction of the amine in the reactive, basic form with the pK_a of the amine at a fixed pH. The observed lack of sensitivity to the rate constants for the nitrosation of secondary amines by N_2O_3[40] and BrNO[52] to their structure addresses the first issue and makes DMA a reasonable model. The second relates to the pK_a of the amine undergoing nitrosation. In studies of the reactivity of secondary amines with nitrosating agents, the observed rate constants have been corrected for the partial protonation of the amine. This was addressed in the simulations by using the pK_a of DMA to calculate the fraction of DMA in the reactive form at the simulation pH (see eq 6). The conclusions drawn from the simulation results will therefore be broadly correct for simple secondary alkylamines that have pK_as similar to DMA. Less basic secondary amines such as morpholine and *N* methylaniline, which have lower pK_as, will nitrosate more rapidly at a given pH

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Table 6. Comparison of the Calculated Initial Rates of Secondary Amine Nitrosation as a Function of Amine pK_a at 25 °C and pH 3.15 Based on Equations 6−8 for a Range of Secondary Amines

amine	pK_a	10^{17} initial rate S1[a] ($M^{-1}\ s^{-1}$)	relative rate S1	10^{6} initial rate S3[b] ($M^{-1}\ s^{-1}$)	relative rate S3
diisopropylamine	11.20[c]	2.86	0.51	0.87	0.51
diethylamine	10.93[d]	5.32	0.96	1.62	0.96
dimethylamine	10.91[e]	5.57	1	1.69	1
ethylisopropylamine	10.76[f]	7.87	1.4	2.39	1.4
4-methylaminobutyric acid	10.66[f]	9.91	1.8	3.01	1.8
morpholine	8.49[c]	1470	264	446	264
N-methylaniline[g]	4.7[f]	8 790 000	1 580 000	2 670 000	1 580 000

[a]S1 = scenario 1: $[NO_2^-] = 2.2 \times 10^{-7}$. M, [amine] = 1 mM, $[Cl^-]$ = 1 M. [b]S3 = scenario 3: $[NO_2^-]$ = 1 M, [amine] = 1 mM, $[Cl^-]$ = 1 mM. [c]Ref 92. [d]Ref 93. [e]Ref 83. [f]Ref 94. [g]The stated initial and relative rates will be an underestimate for the reasons discussed.

because a greater proportion of the amine is in the reactive free base form. Weakly basic secondary amines (p$K_a \leq 9.5$) may therefore represent a higher risk of nitrosation than basic (pK_a > 9.5) secondary amines such as DMA and diethylamine. This is illustrated in Table 6, which compares the calculated initial rates of nitrosation based on eqs 6−8 for a range of secondary alkylamines. *N* Methylaniline and other anilines are likely to react faster than the models discussed in this work predict because the models are based on the rate constants and observed kinetics of secondary amine nitrosation, while the rate constants for the nitrosation of *N* methylaniline by ClNO are approx imately 100 times greater than those reported for DMA and close to those reported for aniline.[54]

Temperature Dependence. The simulations undertaken have used a reaction temperature of 25 °C, as the selection of this temperature gave the most complete set of rate and equilibrium constants. This temperature is higher than the 0−5 °C often used in nitrosation and diazotization reactions and lower than the temperatures at which aqueous washes are sometimes conducted, meaning that it would be useful to be able to make predictions over a range of temperatures. To do this quantitatively based on eqs 6−8 requires the temperature dependence of the constants in the model to be known.

Limited studies of the temperature dependence of the nitrosation of secondary amines by N_2O_3 have been under taken,[47] which show that the enthalpies of activation for a range of secondary alkylamines are low, as expected for a diffusion controlled process, with a range of 10−20 kJ/mol. Based on this range, an activation energy of approximately 17.5 kJ/mol may provide a reasonable upper bound on the extent to which $k_{N_2O_3}$ varies with temperature.[95] By using this value in the variant of the Arrhenius equation that uses a reference rate constant and reference temperature (eq 11), it is possible to estimate that $k_{N_2O_3}$ will increase to $2.1 \times 10^8\ M^{-1}\ s^{-1}$ at 50 °C from $1.2 \times 10^8\ M^{-1}\ s^{-1}$ at 25 °C. The 1.7 fold increase in rate constant predicted for a 25 °C increase in temperature is low relative to the greater than 5 fold increase expected based on the usual doubling of a rate constant for a 10 °C increase in temperature. However, there is no temperature dependence data for the nitrosation of secondary amines by nitrosyl chloride or the nitrous acidium ion, meaning that it is not possible to correct the values of k_{ClNO} and $k_{H_2NO_2^+}$ used in eqs 7 and 8. Based on the assertions in the literature that both[49,53] of these reactions are occurring at close to the limit for a diffusion controlled process, it is reasonable to expect that their rate constants will be similarly insensitive to the reaction temperature

$$k = k_{ref}\ e^{-(E_a/R)((1/T)-(1/T_{ref}))}$$

$$k_{N_2O_3\,T} = 1.2 \times 10^8\ e^{-(17\,500/R)((1/T)-(1/298))} \tag{11}$$

Some temperature dependence data exists for the equilibrium constants for the formation of nitrosyl chloride[44] and nitrosyl bromide[45] (see Table 5), which shows the formation constants to increase with temperature. Similar data does not exist for the formation of N_2O_3.

Finally, the temperature dependence of the dissociation constants of the amine and nitrous acid also needs to be considered. The variation of the pK_as of DMA,[96,97] nitrous acid,[28] and trimethylamine[96,97] with temperature is compared in Table 7. Nitrous acid's pK_a is relatively invariant with temperature over the range studied. More significantly, the pK_as of DMA and trimethylamine decrease significantly with temperature in common with other secondary and tertiary alkylamines. The approximately 0.8 unit fall in the pK_a of DMA on going from 25 to 50 °C equates to a roughly 6 fold increase in K_a. Consequently, the rate of DMA nitrosation will be increased approximately 6 fold at 50 °C due to more of the DMA being present in the reactive basic form.

Considering that the understanding of the temperature dependence is incomplete the model's quantitative predictions cannot simply be extended to other temperatures by including terms to describe the temperature dependence of the constants. However, the available data does suggest that the rate of nitrosation will increase with temperature. It is therefore possible to conclude that if the model described predicts that there is no significant risk of dialkyl *N* nitrosamine formation at 25 °C, there will not be a significant risk at lower temperatures. The effect of the 0.3 unit decrease in the pK_a of DMA for a 10 °C increase in temperature equates to a 2 fold increase in the rate of nitrosation, while the temperature dependence of $k_{N_2O_3}$ suggests that it will increase by a factor of 1.25 if the temperature is

Table 7. Comparison of the Variation of the pK_as of Nitrous Acid, Dimethylamine, and Trimethylamine with Temperature

temperature (°C)	HNO_2[a]	Me_2NH[b]	Me_3N[b]
0		11.55	10.35
10		11.23	10.13
15	3.224		
20	3.177	10.92	9.91
25	3.138	10.91[c]	
30	3.100	10.63	9.69
40		10.35	9.48
50		10.09	9.27

[a]Ref 28. [b]Refs 96 and 97. [c]Ref 83.

Table 8. Predicted Quantities of NDMA Produced in 24 h in a Process Using 10 Relative Volumes of 1 M NaCl for Scenarios 1−3 as a Function of pH and Temperature[a]

Scenario	[DMA]	Nitrite level	Temp. (°C)	ppb of NDMA produced at reaction pH			
				3.15	5	7	9
1a	1 mM[b]	0.01 mg/L[c]	**25**[e]	**3.6×10^{-3}**	**9.9×10^{-5}**	**1×10^{-6}**	**1×10^{-8}**
			35[f]	3.6×10^{-2}	9.9×10^{-4}	1×10^{-5}	1×10^{-7}
			45[f]	3.6×10^{-1}	9.9×10^{-3}	1×10^{-4}	1×10^{-6}
			55[f]	3.6	9.9×10^{-2}	1×10^{-3}	1×10^{-5}
1b	1 mM[b]	3 mg/L[d]	**25**[e]	**1.5**	**5.3×10^{-2}**	**5.3×10^{-4}**	**5.3×10^{-6}**
			35[f]	14.7	5.3×10^{-1}	5.3×10^{-3}	5.3×10^{-5}
			45[f]	147	5.3	5.3×10^{-2}	5.3×10^{-4}
			55[f]	1440	53	5.3×10^{-1}	5.3×10^{-3}
2a	1 M	0.01 mg/L[c]	**25**[e]	**3.5**	**9.9×10^{-2}**	**1×10^{-3}**	**1×10^{-5}**
			35[f]	32	9.9×10^{-1}	1×10^{-2}	1×10^{-4}
			45[f]	145	9.6	1×10^{-1}	1×10^{-3}
			55[f]	163	74	1	1×10^{-2}
2b	1 M	3 mg/L[d]	**25**[e]	**1450**	**53**	**5.3×10^{-1}**	**5.3×10^{-3}**
			35[f]	12300	521	5.3	5.3×10^{-2}
			45[f]	44200	4870	53	5.3×10^{-1}
			55[f]	48200	28900	530	5.3
3	1 mM[b]	1 M	**25**[e]	**740000**	**740000**	**54400**	**560**
			25[e] (1 hour)	740000	210000	2500	25

[a]Key: risk of significant nitrosamine formation (NDMA) based on the lifetime exposure for a 1000 mg daily dose with an acceptable intake (AI) of 96 ng/day green = negligible risk, amber = level >30% of AI potential risk, red = level >AI significant risk. [b]1 mM corresponds to 0.045% w/w relative to the limiting reagent based on a process dilution of 10 relative volumes (10 L/kg). [c]0.01 mg/L nitrite concentrate corresponds to the typical maximum observed level for potable water and equates to a concentration of 2.2×10^{-7} M, which gives a maximum yield of 163 ppb NDMA. [d]3 mg/L nitrite concentration corresponds to the WHO guideline limit for potable water and equates to a concentration of 6.5×10^{-5} M, which gives a maximum yield of 48 200 ppb NDMA. [e]Quantitative predictions based on the published rate and equilibrium constants. [f]Semiquantitative predictions based on the rates of nitrosation increasing by a factor of 10 for a 10 °C temperature increase.

increased by 10 °C. Combining these effects suggests that the rate of nitrosation by N_2O_3 will increase by a factor of 2.5 for a 10 °C temperature increase, provided the temperature dependence of the formation constant, $K_{N_2O_3}$, is negligible. The missing rate constant temperature dependence data mean that a similar

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prediction cannot be made for reaction through ClNO or $H_2NO_2^+$, although the effect is expected to be similar to that observed in the case of N_2O_3. To facilitate some level of prediction, a safety multiplier of 4 was added to the factor of 2.5 predicted for N_2O_3 due to the uncertainty introduced by the missing temperature dependence data and used to provide a worst case prediction for the effect of increasing the temperature by 10 °C. A semiquantitative assessment of the effect of increasing the temperature was therefore made based on a 10 fold increase in rate for a 10 °C increase in temperature, which will overestimate the effect. Additionally, it is also expected that the decomposition of nitrous acid will be accelerated by increasing the reaction temperature, which will have the effect of slowing the rate of amine nitrosation by reducing the nitrous acid concentration. On this basis, it was possible to estimate the level of NDMA that would be formed under scenarios 1−3 at 35, 45, and 55 °C. The resultant values are compared in Table 8.

Based on Table 8, it is apparent that the risk of significant levels of NDMA being formed falls as the pH is increased. It is also clear that scenario 1, trace nitrite in water encountering traces of a basic secondary amine, does not pose a risk in respect of the formation of dialkyl *N* nitrosamines for the nitrite levels typically found in potable water (<0.01 mg/L) at temperatures up to 55 °C. The picture regarding scenario 2, which considers traces of nitrite in water encountering a synthetic stream that utilizes or synthesizes a basic secondary alkylamine, will depend upon the specifics of the process. Water containing <0.01 mg/L nitrite does not pose a significant risk of dialkyl *N* nitrosamine formation at pH values greater than 7 at temperatures below 55 °C. At lower temperatures, a lower pH can be tolerated before there is a significant risk of nitrosamine formation. Regardless of the water quality used, the nitrosation risk due to trace nitrite in water is negligible for processes with a pH greater than 7 when operated at or below 35 °C.

The simulations carried out in this work have shown that in some circumstances it could be possible for trace nitrite in process water to give rise to levels of dialkyl *N* nitrosamines that would exceed the permitted exposure limits if they were to be present in the final isolated API. However, due to the purge[90] that may be provided by the downstream processing steps, the isolated API will not necessarily be contaminated. A thorough assessment of the level of purge provided by the process will be required to determine if a real risk exists.[98,99]

Experimental studies to validate the predictions made based on the models described in this work are currently underway and will be reported in a subsequent publication.

Tertiary Amine Nitrosation. The limited kinetic data that is available for the nitrosation of tertiary amines gives a rate constant of 1.83×10^7 M^{-2} s^{-1} at 75 °C for the nitrosation of triethylamine in 60% aqueous acetic acid with an activation energy of 84.8 kJ/mol.[26] These experiments were conducted via an initial rate methodology at high concentrations of nitrous acid (0.1−0.5 M). Extrapolation using the Arrhenius equation (eq 11) with the reported rate constant as the reference estimates that the rate constant will be 1.34×10^5 M^{-2} s^{-1} at 25 °C. Using this value in the observed rate law (eq 12), it was possible to estimate an initial rate of triethylamine (1 M) nitrosation by nitrous acid (0.1 M) at pH 3.8 and 25 °C. This gave an initial rate of 4.4×10^{-8} M s^{-1}, which is a factor of 200 slower than the initial rate of 9.9×10^{-6} M s^{-1} predicted for the nitrosation of diethylamine under the same concentration conditions based on eqs 6−8. This should be viewed as a lower limit on the difference in the rate between secondary and tertiary amine nitrosation as the use of a solvent system containing significant quantities of acetic acid is likely to have led to the formation of nitrosyl acetate, which can act as a nitrosation catalyst.[100] In contrast, the work of Mirvish suggests that the dealkylative nitrosation of trimethylamine is 8500 times slower than the nitrosation of DMA.[101] These limited data suggest that it is reasonable to conclude that the nitrosation of simple basic tertiary amine nitrosation is likely to be at least 1000 times slower than the nitrosation of the corresponding secondary amine

$$\text{rate} = k_{\mathrm{Et_3N}}[\mathrm{Et_3N}]_T \cdot f_{\mathrm{Et_3N}}[\mathrm{HNO_2}]_T \cdot f_{\mathrm{HNO_2}}[\mathrm{H^+}] \qquad (12)$$

The nitrosation of simple, basic tertiary amines to give dialkyl *N* nitrosamines is therefore slower than the nitrosation of the corresponding secondary amine under the same conditions. This supports the conclusion that there will be no risk of significant dialkyl *N* nitrosamine formation from tertiary amines under conditions where there is no significant risk of dialkylamine nitrosation. However, the rate of reaction under nitrosating conditions is not insignificant and the addition of simple tertiary alkylamines to such reactions must be considered as a potential source of the derived nitrosamines.

■ SUMMARY

The levels of nitrite that may be present in potable water used during API processing are very low, typically <0.01 mg/L. Simulations based on the published kinetics of secondary amine nitrosation have shown that there is no risk that significant levels of dialkyl *N* nitrosamines will be formed by traces of basic secondary amines ($pK_a > 9.5$) encountering water containing this level of nitrite at temperatures below 55 °C. Synthetic concentrations of basic secondary amines are also not expected to form significant levels of dialkyl *N* nitrosamines through reaction with trace nitrite at temperatures up to 55 °C provided the pH is 7 or above. Higher nitrite levels, up to the WHO limit of 3 mg/L, may, in some circumstances, for example, at low pH values or elevated temperatures, give rise to significant levels of *N* nitrosamines. In these cases, and in the case of less basic amines, the models described herein may be used to carry out process specific simulations to predict if there is a real risk. Where significant levels of dialkyl *N* nitrosamines could be formed, the ability of the downstream process operations to purge the *N* nitrosamine may provide sufficient control that there is no risk of the API being contaminated with significant levels of the *N* nitrosamine. Tertiary amine nitrosation can occur through a dealkylative process that is significantly slower than secondary amine nitrosation. There is therefore no risk of basic tertiary alkylamines undergoing significant nitrosation under conditions where secondary amine nitrosation does not represent a risk. The addition of secondary or tertiary alkylamines to synthetic nitrosating conditions should be avoided as this may give rise to significant levels of dialkyl *N* nitrosamines.

■ EXPERIMENTAL SECTION

Kinetic Simulations. Initial rates of nitrosation were calculated as a function of pH in Microsoft Excel using the reactant concentrations stated in the text and the kinetic and equilibrium constants listed in Table 9. An example calculation may be found in the SI.[89]

Kinetic simulations were carried out using Berkeley Madonna version 8.3[102] to numerically solve the differential equations

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Table 9. Summary of the Rate and Equilibrium Constants Used to Calculate Initial Rates of DMA Nitrosation at 25 °C and in Kinetic Simulations of the Nitrosation of DMA

constant	used in equation	value	reference
$k_{N_2O_3}$	6	1.2×10^{8} M^{-1} s^{-1}	47
$k_{H_2NO_2^+}$	7	7000 M^{-2} s^{-1}	49
k_{ClNO}	8	3.1×10^{7} M^{-1} s^{-1}	52
$K_{N_2O_3}$	6	3×10^{-3} M^{-1}	41
K_{ClNO}	8	1.1×10^{-3} M^{-2}	44
pK_a HNO_2	6, 7, 8	3.15	28
pK_a DMA	6, 7, 8	10.91	83

(eqs 6−8) over the desired time period. A copy of the Berkeley Madonna model file in text format may be found in the SI.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.oprd.0c00224.

Sample calculations for converting NDMA concentra tions and conversions into ppb (Figures S1−S3), NDMA formation calculation example for Scenario 1a using eq 10, and Berkeley Madonna model used to simulate NDMA formation (PDF)

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(85) To help put these levels into context, a DMA concentration of 1 mM corresponds to 0.045% (w/w) relative to the limiting reagent based on a process dilution of 10 relative volumes (10 L/kg) or 0.0045% based on a process dilution of 1 relative volume.

(86) The simulated reaction profile for this scenario is contained in Figure S1, SI.

(87) A copy for the initial rate vs pH plot for this scenario with low (1 mM) chloride is contained in Figure S2, SI.

(88) A comparison of the simulated reaction profiles for scenarios 1 and 2 is contained in Figure S3, SI.

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